

Adapted Tree-Sliced Wasserstein: a path-only estimator for adapted optimal transport, with a machine-checked consistency proof

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Abstract

Adapted (bicausal) Wasserstein distances respect the information flow of a stochastic process, but every published algorithm for computing them either fixes a scenario tree in advance or requires access to the transition kernel. We introduce *Adapted Tree-Sliced Wasserstein* (ATSW), an estimator that works from *sampled paths alone*: it quantises paths on a fixed grid, conditions on a window of the last k steps, replaces the optimal-transport problem at each node pair by a closed-form comonotone coupling, and averages over random grid shifts. On the scenario trees used by prior work ATSW is three to four orders of magnitude cheaper than exact dynamic programming, at the cost of a coupling-induced bias that we measure and do not yet bound. Our main negative finding is that a path-only estimator conditioning on the *full* history is statistically infeasible at depth: its bias decays like $n^{-0.19}$, so five per cent accuracy would need on the order of 10^9 paths. Conditioning on a k -step window removes this: on a benchmark with analytic ground truth the bias falls from +42.4% to +1.24%, below a model-based competitor that uses strictly more information. We prove that the estimator is consistent, and the proof is *machine-checked in Lean 4*; formalisation forced three assumptions into the open that an informal argument would have left implicit. We also show that the estimator’s limit law is *not* Gaussian at parameter values where two c.d.f.s coincide, so naive normal confidence intervals are mis-centred rather than merely mis-scaled.

Keywords. adapted Wasserstein distance; bicausal optimal transport; nested distance; comonotone coupling; formal verification.

MSC 2020. [to be assigned]

1 Introduction

1.1 Information flow is not captured by Wasserstein distance

Let μ, ν be laws of discrete-time processes on $\mathbb{R}^T$. The Wasserstein distance $W(\mu, \nu)$ treats a path as a point in $\mathbb{R}^T$ and therefore cannot see *when* information becomes available. The standard two-period example makes this concrete. Let μ reveal a signal of size ε at time 1 and then move to ± 1 determined by that signal; let ν reveal nothing at time 1 and then move to ± 1 by a fair coin. As $\varepsilon \rightarrow 0$ the two laws become indistinguishable in W , yet they are entirely different as filtered objects: under μ a decision maker knows the terminal value one step early, under ν she never does. The adapted distance separates them by $1 + \varepsilon$ while W reports ε .

This matters wherever decisions are sequential: multistage stochastic optimisation, scenario tree generation and reduction, evaluation of generative models for time series, and reward shaping from expert trajectories.

1.2 Computation is the obstacle

Exact adapted transport is a dynamic program over pairs of nodes of two scenario trees. The published methods fall into two families, and *neither* matches the situation a practitioner most often faces — a finite collection of sampled paths and nothing else.

- **Tree-given** methods take two scenario trees as input and solve the nested recursion, exactly or with entropic regularisation. They say nothing about where the tree came from.
- **Model-given** methods sample successor states on demand, which requires the transition kernel. Fitted value iteration is of this type: its inner loop draws successors from the true kernel.

Table 1 collects the published operating points. No entry reports both a large horizon and a large sample size.

1.3 Contributions

1. **Readout, not structure, determines the recovered topology** (Section 4.1). A time-first tree is necessary but not sufficient: on the same tree, a flat tree-Wasserstein readout converges to the wrong limit, a single lexicographic sweep collapses to static Wasserstein, and only a nested readout recovers the adapted distance.
2. **The ATSW estimator** (Section 4): fixed-grid quantisation, k -step conditioning, closed-form comonotone couplings, and averaging over grid shifts. Its cost is output-sensitive.
3. **An informational, not algorithmic, limit** (Section 6.2). With paths only and full-history conditioning the bias decays like $n^{-0.19}$. Two natural repairs (coarsening with depth; data-adaptive cells) both fail, and we explain why.
4. **Consistency, machine-checked** (Section 5). A Lean 4 development with no **sorry**, depending only on Lean’s three standard axioms. Formalisation forced three hypotheses into the open.
5. **A non-Gaussian limit law at ties** (Section 5.3), with a minimal example, a hand derivation, and a simulation that rejects normality.

1.4 Organisation

Section 2 positions the work. Section 3 fixes notation. Section 4 defines the estimator, Section 5 states and proves what we can prove, Section 6 reports experiments, Section 7 the demonstration application, and Section 8 discusses limitations.

Table 1: Published operating points for adapted transport, as reported by their authors on their own hardware. Times are not comparable across rows; the table shows *reach*, not a ranking. Our own entry is a cost milestone only — see Section 6.2 for why it is not an accuracy milestone.

method	source	T	n	time (s)	note
Nested distance, exact (Pichler)	2102.05413	5	144	10.29	tree 144x24 leaves
Nested Sinkhorn (same paper)	2102.05413	5	144	2.87	same instance
Adapted empirical measure, aggregated	EP 2022	3	10000	14.38	1 CPU 3.1 GHz
Adapted empirical measure, non-aggregated	EP 2022	3	2000	—	reported as stopping here
Fitted value iteration	FVI 2023	40	4000	257.70	i5 + RTX 3050 Ti
Backward-induction LP	FVI 2023	11	2048	794.00	reported baseline
Adapted Sinkhorn	FVI 2023	10	1024	1443.60	reported baseline
KL-relaxed adapted OT	2026	10	2000	31.60	interpolated value
ATSW (this work)	—	50	10000	6.60	laptop CPU, cost only

2 Related work

2.1 Theory of adapted transport

The adapted (nested) distance was introduced to make multistage stochastic programs stable under perturbation of the underlying process. On the real line and for monotone costs the Knothe–Rosenblatt coupling is optimal among causal couplings, which is what makes a quantile-based surrogate plausible in the first place. A separate line studies *adapted empirical measures* — how fast an empirical process converges in adapted distance — and it is this line that our consistency result speaks to.

2.2 Computation of adapted transport

2.3 Sliced and tree-sliced transport

A large literature obtains $O(Ln \log n)$ transport by projecting to one dimension or onto a tree metric, where the optimal coupling is available in closed form. It is worth stating explicitly why these are not competitors here: their speed comes precisely from the absence of a filtration to respect. The contribution of the present work is to keep a closed-form coupling *inside* a nested recursion, which is where the difficulty lies.

3 Preliminaries

3.1 Notation

For $x \in \mathbb{R}^T$ write $x_{1:t}$ for the first t coordinates. All processes take values in a finite set after quantisation; n and m denote the numbers of cells for the two processes, B a bound on the cost matrix, and $\|\cdot\|_1$ the ℓ^1 norm on mass vectors.

[figure file missing — see the caption below]

Figure 1: The same non-anticipative tree read three ways. Only the nested readout recovers the adapted distance.

3.2 Bicausal couplings and the adapted distance

Definition 1. A coupling π of $\mu, \nu \in \mathcal{P}(\mathbb{R}^T)$ is *bicausal* if, under π , $Y_{1:t}$ is conditionally independent of $X_{t+1:T}$ given $X_{1:t}$, and symmetrically. Write $\text{Cpl}_{bc}(\mu, \nu)$ for the set of such couplings and

$$\mathcal{AW}_p^p(\mu, \nu) = \inf_{\pi \in \text{Cpl}_{bc}(\mu, \nu)} \int \sum_{t=1}^T |x_t - y_t|^p d\pi.$$

3.3 The nested recursion

Bicausality is what makes the problem decompose. With $V_T \equiv 0$,

$$V_t(x_{1:t}, y_{1:t}) = \inf_{\pi_t \in \text{Cpl}(\mu_{t+1|x_{1:t}}, \nu_{t+1|y_{1:t}})} \int [c_{t+1}(x_{t+1}, y_{t+1}) + V_{t+1}] d\pi_t,$$

and $\mathcal{AW}_p^p = V_0$. Every algorithm in this area is a way of doing something cheaper at the inner inf, or of visiting fewer $(x_{1:t}, y_{1:t})$ pairs. ATSW does both.

3.4 Comonotone couplings

For probability vectors p, q on ordered finite supports with c.d.f.s F, G , the comonotone coupling assigns mass

$$\ell_{ij}(p, q) = (\min(F_i, G_j) - \max(F_{i-1}, G_{j-1}))^+. \quad (1)$$

It costs $O(b \log b)$ for supports of size b , and is optimal in one dimension for monotone costs. Two properties of (1) drive everything later: it is piecewise linear in (p, q) , and it is built from min and max, hence only *directionally* differentiable — the source of the non-Gaussian limit in Section 5.3.

4 The ATSW estimator

4.1 Readout determines the recovered topology

A natural first attempt is to build a time-first tree and apply an existing tree-Wasserstein readout. On the two-period example of the introduction, whose adapted distance is $1 + \varepsilon = 1.30$ at $\varepsilon = 0.3$, we measured three readouts on the *identical* tree as the cell width is refined:

Algorithm 1 ATSW at one grid shift

Require: paths $x^{(1..n)}, y^{(1..n')} \in \mathbb{R}^T$; grid width δ ; shift s ; window k ; cost c

- 1: $C_t^x \leftarrow \lfloor (x_t - s)/\delta \rfloor$, representative = cell centre $\triangleright$ deterministic given (δ, s)
 - 2: group paths by the tuple of their last k cell indices at each time $\triangleright$ node = window state
 - 3: $V \leftarrow 0$
 - 4: **for** $t = T - 1$ **down to** 0 **do**
 - 5: **for** each pair of window states (u, v) carrying mass **do**
 - 6: $V(u, v) \leftarrow \text{COMONOTONECOST}(\hat{K}^x(u, \cdot), \hat{K}^y(v, \cdot), c + V)$ $\triangleright$ Eq. (1)
 - 7: **end for**
 - 8: **end for**
 - 9: **return** V at the root
-

- a flat tree-Wasserstein readout (ℓ^1 on cylinder-cell masses) converges to 4.0;
- a single lexicographic sweep returns ε , the static Wasserstein value;
- the nested readout returns $1 + \varepsilon$ exactly.

The mechanism of the lexicographic failure is that it matches marginal mass before conditioning, and conditioning does not commute with coupling. The conclusion is structural: a time-first tree is necessary but not sufficient, and the nested readout is not an implementation detail.

4.2 Construction

Three choices in Algorithm 1 deserve comment.

Fixed grid rather than data-adaptive cells. Cell boundaries and representatives are deterministic given (δ, s) . Cells built from sample quantiles are more accurate in some regimes (Section 6.4), but they make the cell boundaries random, and then the consistency proof of Section 5 no longer covers the estimator that is actually run. We chose the estimator the theorem covers.

Window of length k . Conditioning on the last k cell indices bounds the number of nodes by $(\text{number of cells})^k$, independent of T . Section 6.3 shows k is a two-sided knob with an interior optimum.

Fallback on unvisited cells. If a source cell is never visited the count ratio is 0/0. The estimator returns the uniform row there. This is not cosmetic: it is what makes every estimated row a probability vector at every sample size, which the consistency proof requires (Remark 5).

4.3 Cost

The number of visited segments grows like $n^{1.24}$ at $T = 10$ and $n^{1.34}$ at $T = 50$ (Table 2), i.e. sub-quadratic but super-linear, and the exponent itself drifts upward with depth. A configuration with $n = 10^5$, $T = 100$ completes in about ten minutes of laptop CPU. *This is a cost milestone and not an accuracy milestone*; Section 6.2 explains why the two must be reported separately.

Table 2: Segments visited by the recursion. The cost is output-sensitive: only node pairs carrying mass are visited. The fitted exponent within each depth is given in the last column.

T	n	segments visited	time (s)	fitted exponent in n
10	250	15921	0.335	1.235
10	500	38203	0.820	1.235
10	1000	89782	1.925	1.235
10	2000	207620	4.464	1.235
50	250	106708	2.143	1.341
50	500	280253	5.677	1.341
50	1000	696908	13.808	1.341
50	2000	1744941	33.686	1.341

5 Theory

5.1 Consistency

Let $\hat{K}_M$ be the count-based kernel estimator from M observed transition pairs with the uniform fallback row, and K^0 the true kernel.

Assumption 2 (Reachability). Every source cell has strictly positive probability under the law of the first coordinate of a transition pair.

Assumption 3 (Sampling). The pooled transition pairs are pairwise independent, identically distributed and measurable.

Theorem 4 (Consistency). *Under Assumptions 2 and 3 and a bounded cost, the backward-recursion value computed from $\hat{K}_M$ converges almost surely, as $M \rightarrow \infty$, to the value computed from K^0 .*

Proof sketch. The argument is a continuous-mapping argument with an explicit modulus. (i) The c.d.f. is 1-Lipschitz in the mass vector, hence by (1) each comonotone weight is Lipschitz in (p, q) with constant 2; (ii) since the coupling cost is linear in the cost matrix and the weights are bounded, $|QC(p, q; M) - QC(p', q'; M)| \leq 2nmB(\|p - p'\|_1 + \|q - q'\|_1)$; (iii) one recursion step is jointly Lipschitz in the kernel and in the continuation value; (iv) composing s steps gives a Lipschitz bound with an explicit, recursively defined constant; (v) a Lipschitz bound whose input vanishes yields convergence of the value; (vi) the strong law of large numbers gives almost-sure convergence of empirical frequencies, and under Assumption 2 the ratio of frequencies converges to the true kernel. $\square$

5.2 Machine-checked formalisation

The proof above is formalised in Lean 4 with mathlib. The development is **35 results, 613 lines, zero sorry**, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for every statement, including Theorem 4.

Remark 5 (What formalisation forced into the open). Three hypotheses that an informal proof leaves implicit had to be stated. (i) *Reachability*. Without Assumption 2 the estimated kernel is undefined on unvisited cells and the ratio has no limit. (ii) *The fallback row*. For finite M the denominator can vanish, so the estimator must return a genuine probability vector in that case; otherwise the hypothesis of the composition step fails at small M and the induction does not start.

Table 3: The Lean development. The final row is Theorem 4.

name	statement
<code>cdf_lipschitz</code>	the c.d.f. is 1-Lipschitz in the mass vector
<code>ell_nonneg, ell_le_one</code>	comonotone weights lie in $[0, 1]$
<code>ell_lipschitz</code>	each weight is Lipschitz in (p, q)
<code>QC_linear_M</code>	the coupling cost is linear in the cost matrix
<code>QC_lipschitz</code>	$ QC(p, q; M) - QC(p', q'; M) \leq 2nmB(\ p - p'\ _1 + \ q - q'\ _1)$
<code>QC_step_lipschitz</code>	one recursion step, jointly in kernel and continuation
<code>Vdp_bound, Vdp_lipschitz</code>	the T -step value, with explicit constants
<code>Vdp_tendsto</code>	vanishing kernel error implies convergence of the value
<code>empirical_freq_ae</code>	strong law for empirical frequencies
<code>sum_measure_real, K0_isProb</code>	true kernel rows are probability vectors
<code>Khat_isProb</code>	estimated rows are probability vectors for every M
<code>Khat_ae_tendsto</code>	the count estimator converges a.s. under reachability
<code>Vdp_consistent_ae</code>	a.s. convergence of the estimated value

Convergence is unaffected because the denominator is eventually nonzero. *(iii) Marginal equals sum of joints.* Needed to show the true kernel rows are probability vectors; formally this is an integral identity, not an immediate rewriting.

Remark 6 (Scope of the formal statement). The Lipschitz constant grows like $(nm)^s$ and therefore certifies *continuity*, not a *rate*. The formal statement is for a time-homogeneous kernel — the stationary k -Markov case used throughout — and models the pooled transition pairs as i.i.d.

5.3 A non-Gaussian limit at ties

Proposition 7. *Let $p = (0.2, 0.5, 0.3)$ and $q = (0.2, 0.6, 0.2)$ on $\{0, 1, 2\}$ with cost $|x - y|$, so that the two c.d.f.s coincide at the first cut point. Let $\hat{V}$ be the comonotone cost between independent empirical versions of p and q . Then $\sqrt{N}(\hat{V} - V)$ converges to $|A| + B$ for jointly Gaussian A, B , whose mean is 0.4514 and is in particular nonzero.*

The mechanism is that (1) is directionally but not fully differentiable: where two c.d.f.s coincide, the min/max structure makes the first-order expansion a nonlinear function of the Gaussian fluctuation. Simulation with $R = 200,000$ replicates at $N = 50,000$ gives mean 0.4504 (hand derivation 0.4514), skewness 0.239, and a Kolmogorov–Smirnov test against the centred Gaussian with $p < 10^{-12}$.

The practical consequence is sharper than “the limit is non-normal”: a confidence interval built on a naive normal approximation is *mis-centred*, not merely mis-scaled, and no choice of variance estimator repairs it.

5.4 Rate of convergence

NOT YET WRITTEN.

Theorem 4 gives no rate, and Remark 6 explains why the constant we prove cannot yield one. The target statement is a bound of the form $\mathbb{E}|\hat{V}_M - V| \lesssim \rho(n, m, k, T) M^{-1/2}$ under Assumption 2 with a quantitative lower bound on cell probabilities, obtained by replacing step (iv) of the proof

[figure file missing — see the caption below]

Figure 2: Limit law of $\sqrt{N}(\hat{V} - V)$ at a tie. The centred Gaussian that a naive delta method would predict is displaced from the truth by the mean of the true limit.

sketch with a martingale argument rather than a worst-case composition. What would falsify it: the measured bias in Section 6.2 decays like $n^{-0.19}$, far slower than $M^{-1/2}$, so any claimed rate must be stated for the kernel error and not for the path count, and the two must be related through the number of realisable window states.

5.5 Bounding the comonotone-coupling gap

NOT YET WRITTEN.

Section 6.1 measures a 10–20% gap between the comonotone surrogate and the exact inner optimum on multi-branch trees. We have no bound on it. The target is a bound in terms of branching factor and of the departure of the conditional cost from monotonicity, recovering exactness in the one-dimensional monotone case. What would falsify it: an instance family on which the ratio grows with branching factor without bound.

5.6 Metric properties

NOT YET WRITTEN.

Whether the ATSW functional is a metric on the space of quantised laws — symmetry is clear, the triangle inequality is not — and whether the shift-averaged version metrises the same topology as $\mathcal{AW}$ on a suitable class.

[figure file missing — see the caption below]

Figure 3: Head-to-head on the competitor’s own instance generator.

Table 4: Scenario trees from the competitor’s generator; n_b is the branching parameter, times in seconds, error relative to exact dynamic programming. The support is integer-valued, so $\delta = 1$ is lossless and the ATSW error column isolates the cost of replacing the linear program by a comonotone coupling.

T	n_b	paths	DPP exact	DPP Sinkhorn	ATSW $\delta=1$	ATSW err. (%)	Sinkhorn err. (%)
2	10	100	0.05	0.35	0.0007	22.6	6.3
2	25	625	1.66	3.72	0.0009	13.9	16.3
2	50	2500	22.96	29.74	0.0024	9.7	29.1
2	75	5625	105.91	119.48	0.0076	14.8	63.8
2	100	10000	258.94	264.25	0.0082	23.4	150.5
3	10	1000	3.22	21.87	0.0021	22.4	2.2

6 Experiments

Protocol. All timings are on the same machine. Where a competitor is compared we run *their* published code on *their* instance generator; a re-implementation is used only to understand mechanism, never to claim a ratio. Ground truth is analytic where available and otherwise exact dynamic programming with a linear-programming inner solver. Every number in the tables is generated from the result files at build time.

6.1 Given a scenario tree

ATSW is three to four orders of magnitude cheaper. The honest counterpart is the error column: replacing the linear program by a comonotone coupling costs roughly ten to twenty per cent on these instances, and we have no bound on that gap.

The entropic branch of the competitor’s own code is not a usable middle ground here. Its documentation states that the method should converge to unregularised transport as $\varepsilon \rightarrow 0$, while warning that values below roughly $\varepsilon = 0.01$ may cause problems in practice; our sweep (Appendix C) is consistent with that warning and shows the practical range to be narrow — the relative error is +16 to +24% at $\varepsilon = 10^{-2}$, passes through -4 to -16% at 10^{-3} , and degrades monotonically to -92 to -96% at 10^{-5} . The iteration breaks down numerically before the regularisation is small

[figure file missing — see the caption below]

Figure 4: With paths only and full-history conditioning the bias decays like $n^{-0.19}$. The competitor samples successor states from the true kernel, i.e. it requires a model.

Table 5: Gaussian benchmark, $T = 8$, analytic ground truth 17.0. FVI samples successors from the true kernel and therefore uses strictly more information than any path-only estimator.

estimator	n	value	bias (%)	time (s)
full history, $\delta = 0.5$	2000	30.391	78.77	0.12
full history, $\delta = 0.5$	10000	27.598	62.34	1.02
full history, $\delta = 0.5$	50000	24.213	42.43	9.75
k -Markov, quantile cells $m = 40$	2000	18.917	11.28	0.32
k -Markov, quantile cells $m = 40$	10000	18.330	7.82	0.55
k -Markov, quantile cells $m = 40$	50000	17.295	1.73	0.92
k -Markov, fixed grid $\delta = 0.125$	50000	17.211	1.24	1.97
FVI, model-based (original code)	2000	17.484	2.85	14.20

enough to be accurate, so we compare against the exact dynamic program.

6.2 Paths only: an informational limit

Full-history conditioning is not merely inaccurate, it is inaccurate in a way that more data does not fix: fitting the three measured points gives a bias decaying like $n^{-0.19}$, so five per cent would require on the order of 4×10^9 paths. The cause is that the number of realisable histories grows like $(\text{number of cells})^T$, so at depth almost every history is observed once and the conditional laws are estimated from a single sample.

Two natural repairs fail, and their failure is informative. Coarsening the grid with depth is much worse (+460% at the most aggressive setting) because coarse cell centres introduce a cost error that dominates the sampling gain. Data-adaptive quantile cells do not help either, because they change *where* the cells are but not *how many* histories there are.

6.3 Window length is a two-sided knob

The benchmark is deliberately non-Markov — transition probabilities depend on the whole history — so that a window that is too short must lose information. It does, and it biases *downwards*,

[figure file missing — see the caption below]

Figure 5: Conditioning on the last k steps. Too small a window under-conditions and biases downwards; the full history over-fragments the sample and biases upwards.

Table 6: Non-Markov tree, $T = 4$, branching 3, $n = 50000$; ground truth 5.7947 by exact dynamic programming. The window length has an interior optimum.

conditioning	value	bias (%)	time (s)
$k=1$, $m=27$	5.637	-2.73	0.218
$k=2$, $m=27$	5.808	0.23	0.281
$k=3$, $m=27$	5.808	0.23	0.306
full history, $\delta = 0.3$	5.862	1.16	—

in the opposite direction to the over-fragmentation bias. Note that the two longer windows are indistinguishable at this depth, so these data locate the optimum only up to $k \in \{2, 3\}$.

6.4 Fixed grid versus data-adaptive cells

This is a deliberate trade, not an improvement: we accept more bias in the under-conditioned regime in exchange for an estimator the theorem covers. The grid width has an interior optimum for the same sampling reason as the window length — on the Gaussian benchmark $\delta = 0.0625$ is worse than $\delta = 0.125$.

6.5 Selecting k and δ

NOT YET WRITTEN.

Both parameters have interior optima and both are currently chosen by sweep against a ground truth that a user will not have. The target is a selection rule computable from the sample alone — a held-out split on paths, or a stability criterion across grid shifts — validated against the swept optimum on the benchmarks of this section. What would falsify it: a rule whose selected (k, δ) is worse than a fixed default across the benchmark set.

Table 7: Every configuration run. The fixed grid is what Theorem 4 covers. It is worse in 3 of 5 configurations and comparable at the operating point actually used (interior-optimal k , large n).

benchmark	conditioning	quantile cells (%)	fixed grid (%)	better
non-Markov tree $T=4$, $n=5000$	$k=1$	-1.02	-2.68	quantile
non-Markov tree $T=4$, $n=5000$	$k=2$	0.69	1.05	quantile
non-Markov tree $T=4$, $n=50000$	$k=1$	-2.73	-4.17	quantile
non-Markov tree $T=4$, $n=50000$	$k=2$	0.23	0.07	fixed
Gaussian $T=8$, $n=50k$, δ swept	$k=1$	1.73	1.24	fixed

7 Demonstration application

NOT YET WRITTEN.

The experiments above are benchmarks against ground truth, which establishes correctness but not usefulness. One application will be developed in full. Two candidates, both verified to be open in the literature survey:

(A) Evaluation metric for generative time-series models. *Current practice compares marginal or two-point statistics, which are blind to information flow; a generator that leaks future information scores well. Protocol: train several generators on the same data, rank them by ATSW against held-out paths, and check the ranking against a downstream sequential decision task. What would falsify the claim: ATSW ranking no better than a static Wasserstein baseline at predicting downstream performance.*

(B) Reward from expert trajectories in imitation learning. *Existing OT-based rewards use static transport between state-visitation distributions. Protocol: substitute ATSW and measure sample efficiency against the static baseline on standard control tasks. What would falsify the claim: no improvement once the static baseline is given the same amount of tuning.*

Option (A) is closer to the estimator’s guarantees; option (B) has the wider audience.

8 Discussion

8.1 What the negative result means

It is tempting to read Section 6.2 as a defect of ATSW. It is not: any estimator that sees only paths faces the same count of realisable histories, and the model-based competitor beats it precisely by using information ATSW is not given. The useful reading is that adapted transport from paths alone is feasible only under a conditioning restriction, and that choosing the restriction — rather than choosing the coupling — is where the accuracy is decided.

8.2 Limitations

- The comonotone coupling is not optimal on multi-branch trees; on the scenario-tree benchmark this costs ten to twenty per cent, and we have no bound.
- Neither k nor δ has a data-driven selection rule; both have interior optima.

- The consistency constant certifies continuity, not a rate.
- The formal development treats pooled transition pairs as i.i.d.; dependence between pairs drawn from the same path is not treated.
- The benchmarks use one-dimensional state; multivariate state is quantised coordinatewise and has not been evaluated.
- No demonstration application is included yet.

9 Conclusion

Adapted transport from sampled paths is feasible, but the binding constraint is statistical rather than computational, and the remedy is to condition on a bounded window rather than to improve the coupling. The resulting estimator is consistent, and that statement is machine-checked rather than argued.

A Proof details

NOT YET WRITTEN.

A human-readable rendering of the Lean development: the explicit constants in steps (iii) and (iv) of the proof sketch, and the derivation of the limit law in Proposition 7.

B The Lean development

Built with Lean 4.33.0 and mathlib at tag v4.33.0. Reproduce with `lake exe cache get && lake build`; `Check.lean` prints the axiom dependencies of every statement. Importing all of mathlib exhausts 8 GiB, so the file imports six modules selectively.

C Entropic sweep

D Reproducibility

Every number in this paper is generated from result files at build time by `build_paper.py`; no figure or table value is transcribed by hand. The estimator is `atsw_fixedgrid.py` (fixed grid, the version Theorem 4 covers) with `atsw_kmarkov.py` providing the windowed recursion, and the formal development is `Atsw.lean`.

Data availability

[to be completed]

Table 8: Entropic branch of the competitor’s code across regularisation levels, two instance sizes and two seeds.

n_b	seed	ε	exact	Sinkhorn	rel. err. (%)	time (s)
25	0	0.01	0.040929	0.047515	16.1	3.01
25	0	0.001	0.040929	0.034178	-16.5	5.10
25	0	0.0001	0.040929	0.012106	-70.4	2.56
25	0	1e-05	0.040929	0.001713	-95.8	1.43
25	1	0.01	0.029506	0.036489	23.7	2.99
25	1	0.001	0.029506	0.028318	-4.0	5.46
25	1	0.0001	0.029506	0.015622	-47.1	2.90
25	1	1e-05	0.029506	0.002508	-91.5	1.50
50	0	0.01	0.041040	0.049146	19.8	24.98
50	0	0.001	0.041040	0.036870	-10.2	35.18
50	0	0.0001	0.041040	0.010716	-73.9	24.58
50	0	1e-05	0.041040	0.001552	-96.2	19.94
50	1	0.01	0.033887	0.041579	22.7	22.89
50	1	0.001	0.033887	0.031977	-5.6	32.59
50	1	0.0001	0.033887	0.009850	-70.9	22.31
50	1	1e-05	0.033887	0.001516	-95.5	18.29

Acknowledgements

[to be completed]